PROJECT-BASED LEARNING

PHASE-I EVALUATION

EcoLife – Ecosystem Simulation and Analysis System

Team ID: *DSCPP-III-2026-T169*

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: *DSCPP-III-2026-T169*
Semester: *3rd*
Section: *XX*
Domain: DSA (C) & OOP (C++)

Team Members

1. Mishra Adarsh – 2510372901 (Lead)
2. Gupta Unnati – 2510372161
3. Budakoti Rishabh – 25101201427
4. Dheeman Amritaanshu – 2512520011

Mentor

Ms. Shrishti Chamoli

Interactions completed: 2

Problem Statement & Motivation

Problem Statement

An ecosystem is a network of interdependent organisms and environmental conditions, not a set of isolated species. A single disturbance – drought, fire, or the loss of a predator – can cascade through the food web and affect multiple populations over time. Real conditions cannot be safely controlled or repeated for testing in nature.

Motivation

- Real ecosystems cannot be experimented on safely
- Interactions among species are hard to trace by hand
- Existing scientific models are large and not education-friendly

Target Users / Stakeholders

- Students learning ecology, DSA and OOP concepts
- Instructors demonstrating population dynamics
- Anyone exploring what-if ecological scenarios

Objectives

Project Objectives

1. Build a virtual ecosystem where organisms and environmental conditions influence each other over simulated time.
2. Model plants, herbivores, carnivores and birds using a class hierarchy with inheritance and polymorphism.
3. Update populations based on food availability, feeding, predation, reproduction and death.
4. Incorporate weather, seasons and natural events (drought, forest fire, predator removal).
5. Support what-if comparisons between normal and disturbance scenarios.

Key Objective: Simulate how a change in one population propagates through the food web over simulated days.

Proposed Solution

Daily Simulation Cycle

1. Update environmental conditions and resources
2. Process scheduled events (drought, fire, etc.)
3. Apply organism interactions (feeding, predation, reproduction, death)
4. Update population values
5. Present results through a menu-driven interface

Key Features

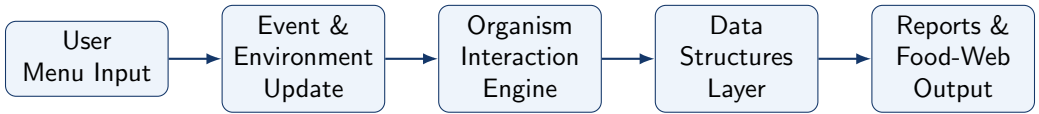
- Advance simulation by 1, 10 or 100 days
- View current population and event history
- Explore food-web relationships (graph traversal)
- Simple ecosystem-health indicator
- What-if testing: drought, fire, predator removal

Uses simple, understandable rules rather than full scientific accuracy, making cause-and-effect relationships easy to observe.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Arrays (population data), linked lists (dynamic organism collections), queues/priority queues (event scheduling), trees & hash tables (organism lookup), graphs with DFS/BFS (food-web traversal), sorting/searching (population analysis)	Event manager, population store, food-web analyser
OOPs with C++	Inheritance (Organism base class extended by Plant, Herbivore, Carnivore, Bird), polymorphism (uniform handling of feeding, predation and reproduction across organism types), composition (Environment built from Weather, Season and Event objects), encapsulation (Ecosystem class controlling overall simulation state)	Organism hierarchy, environment model, ecosystem controller

System Architecture / Workflow



Major Components

- Organism hierarchy (Plant/Herbivore/Carnivore/Bird)
- Environment, Weather, Season, Event modules
- Food-web graph with DFS/BFS analysis

Data / Control Flow

- User advances simulation by N days via menu
- Event queue and environment state update populations
- System reports population counts, events and food-web health

Technology Stack & Methodology

Technology Stack

- Programming: C and C++
- Storage: In-memory data structures (no external DB)
- Tools: GCC / g++, Make
- IDE: *VS Code / Code::Blocks*
- Version Control: Git / GitHub

Development Methodology

1. Requirement Analysis
2. Class & Data-Structure Design
3. Implementation (organism model + DSA modules)
4. Testing (scenario-based: drought, fire, predator removal)
5. Evaluation & Report

C++ is used for the organism/environment model to leverage inheritance and polymorphism; C-style data structures give explicit control over event scheduling and food-web analysis performance.

Initial Progress & Team Contribution

Progress Achieved

- Problem study and motivation finalised
- Organism class hierarchy planned
- Data structures mapped to modules
- Daily simulation cycle designed
- Phase-I proposal and report completed

Individual Contribution

Member	Role	Contribution
Adarsh Mishra	Team Lead	Coordinated team & designed system architecture
Unnati Gupta	C++ Design	Designed `Organism/Plant/Animal` hierarchy
Rishabh Budakoti	DSA Design	Planned event queue, lookup & food-web graph
Amritaanshu Dheeman	Docs & Test Plan	Drafted report & designed test scenarios

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design (current)
2. Phase-II: Organism model, event system & core DSA modules
3. Phase-III: Full simulation, testing & final report

Expected Outcomes

- Working menu-based ecosystem simulator
- Population, event and food-web reports
- What-if scenario testing
- Extendable design for new organism types

References

1. B. Stroustrup, *The C++ Programming Language*, 4th ed., Addison-Wesley, 2013.
2. B. W. Kernighan, D. M. Ritchie, *The C Programming Language*, 2nd ed., Prentice Hall, 1988.
3. Cormen, Leiserson, Rivest, Stein, *Introduction to Algorithms*, 4th ed., MIT Press, 2022.
4. OpenStax, *Biology 2e*, "46.1 Ecology of Ecosystems."
5. OpenStax, *Biology 2e*, "45.6 Community Ecology."

Thank You – Questions & Suggestions